

Exact paper stack: original (A) vs math-layout (D)

bayes.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D} \mid \vartheta) p(\vartheta) d\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta) p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)} [\prod_{i=1}^n p(y_i \mid x_i, \vartheta)]} + \frac{1}{2} \log \left(\frac{|\Sigma_0|}{|\Sigma_{\theta|\mathcal{D}}|} \right) + \text{KL} (q(\theta) \parallel p(\theta)) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta)$$

D: math-layout

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D} \mid \vartheta) p(\vartheta) d\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta) p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)} [\prod_{i=1}^n p(y_i \mid x_i, \vartheta)]} + \frac{1}{2} \log \left(\frac{|\Sigma_0|}{|\Sigma_{\theta|\mathcal{D}}|} \right) + \text{KL} (q(\theta) \parallel p(\theta)) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta) - \frac{1}{2} \log \det \Sigma_{\theta|\mathcal{D}}$$

diffusion.tex eq.0 (continuation, 2 lines)

A: original

$$\mathbb{E} \left[\|x_{t+1} - x_t\|^2 \right] = \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right] + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2} \|\nabla_x \log p_t(x_t) - s_{\theta}(x_t, t)\|^2$$

D: math-layout

$$\mathbb{E} \left[\|x_{t+1} - x_t\|^2 \right] = \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right] + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2} \|\nabla_x \log p_t(x_t) - s_{\theta}(x_t, t)\|^2$$

esr.tex eq.0 (continuation, 2 lines)

A: original

$$\mathcal{L}(\theta) = \frac{1}{2} \log (2\pi \sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_{\theta}(x_i))^2 + \frac{1}{2} \log |\Sigma_{\theta}| + \frac{d}{2} \log (2\pi) + \lambda \sum_{j=1}^p |\theta_j| + \frac{n}{2} \log \left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right)$$

D: math-layout

$$\mathcal{L}(\theta) = \frac{1}{2} \log (2\pi \sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_{\theta}(x_i))^2 + \frac{1}{2} \log |\Sigma_{\theta}| + \frac{d}{2} \log (2\pi) + \lambda \sum_{j=1}^p |\theta_j| + \frac{n}{2} \log \left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right)$$

kalman.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} H_t^{\top} \left(H_t P_{t|t-1} H_t^{\top} + R_t \right)^{-1} H_t P_{t|t-1} + F_t P_{t-1|t-1} F_t^{\top} - K_t S_t K_t^{\top} + \Gamma_t u_t u_t^{\top} \Gamma_t^{\top} - \Phi_t \Sigma_{t|t-1} \Phi_t^{\top} + \Omega_t$$

D: math-layout

$$\begin{aligned} P_{t|t} &= P_{t|t-1} - P_{t|t-1} H_t^\top \left(H_t P_{t|t-1} H_t^\top + R_t \right)^{-1} H_t P_{t|t-1} \\ &\quad + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t \end{aligned}$$

kkt.tex eq.0 (continuation, 2 lines)

A: original

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m (\max\{0, g_i(x)\})^2 + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^*\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho}$$

D: math-layout

$$\begin{aligned} \mathcal{L}(x, \lambda, \nu) &= f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m (\max\{0, g_i(x)\})^2 \\ &\quad + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^*\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho} \end{aligned}$$

koopman.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$\tilde{A} = \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y = (\Psi_X^\top \Psi_X)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon (\Psi_X^\top \Psi_X + \lambda I)^{-1} \Psi_X^\top \Psi_Y + \|\Psi_Y - \Psi_X \tilde{A}\|_F^2 + \lambda \|\tilde{A}\|_* + \|\Psi_X^\top \Psi_Y - \tilde{A}\|_F$$

D: math-layout

$$\begin{aligned} \tilde{A} &= \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y \\ &= (\Psi_X^\top \Psi_X)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon (\Psi_X^\top \Psi_X + \lambda I)^{-1} \Psi_X^\top \Psi_Y + \|\Psi_Y - \Psi_X \tilde{A}\|_F^2 + \lambda \|\tilde{A}\|_* + \|\Psi_X^\top \Psi_Y - \tilde{A}\|_F \end{aligned}$$

lyapunov.tex eq.0 (continuation, 2 lines)

A: original

$$\lambda_{\max} \left(A^\top P + PA + Q \right) \leq -\gamma \|x(t)\|^2 + \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, dt - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, ds - \delta \|x\| \|\dot{x}\|$$

D: math-layout

$$\begin{aligned} \lambda_{\max} \left(A^\top P + PA + Q \right) &\leq -\gamma \|x(t)\|^2 + \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, dt \\ &\quad - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, ds - \delta \|x\| \|\dot{x}\| \end{aligned}$$

matrix.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$P = A (A^\top A + \lambda I)^{-1} A^\top = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} = U_r U_r^\top - (I - U_r U_r^\top) + 2U_r U_r^\top - (I - U_r U_r^\top) + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1}$$

D: math-layout

$$P = A (A^\top A + \lambda I)^{-1} A^\top = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \\ = U_r U_r^\top - (I - U_r U_r^\top) + 2U_r U_r^\top - (I - U_r U_r^\top) + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1}$$

matrix.tex eq.1 (continuation-unanchored, 2 lines)

A: original

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \quad A^\dagger = V_r \Sigma_r^{-1} U_r^\top = \sum_{k=1}^r \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} + \lambda \mathcal{N}(A)^\perp$$

D: math-layout

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \quad A^\dagger = V_r \Sigma_r^{-1} U_r^\top \\ = \sum_{k=1}^r \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} + \lambda \mathcal{N}(A)^\perp$$

nash.tex eq.0 (continuation, 2 lines)

A: original

$$\sigma^* \in \arg \max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j + \eta \sum_{i \in S} \sum_{j \in T} (\sigma_i \tau_j)^2 - \kappa \|\sigma - \sigma^*\|_1 + \gamma \|\nabla_\sigma \mathbb{E}[u]\|^2 - \beta \sum_{i \in S} \sigma_i^2$$

D: math-layout

$$\sigma^* \in \arg \max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j \\ + \eta \sum_{i \in S} \sum_{j \in T} (\sigma_i \tau_j)^2 - \kappa \|\sigma - \sigma^*\|_1 + \gamma \|\nabla_\sigma \mathbb{E}[u]\|^2 - \beta \sum_{i \in S} \sigma_i^2$$

network-flow.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$\sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \quad \forall v \in V, \quad 0 \leq x_a \leq u_a, \quad \forall a \in A, \quad \sum_{v \in V} b_v = 0, \quad \sum_{a \in A} c_a x_a \leq C_{\max}, \quad x_a \geq 0$$

D: math-layout

$$\sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \quad \forall v \in V, \quad 0 \leq x_a \leq u_a, \quad \forall a \\ \in A, \quad \sum_{v \in V} b_v = 0, \quad \sum_{a \in A} c_a x_a \leq C_{\max}, \quad x_a \geq 0$$

pathological.tex eq.1 (single-line, 1 lines)

A: original

$$f\left(g\left(h\left(k\left(x_1+x_2+x_3+x_4+x_5+x_6+x_7\right)\right)\right)\right)+\text{MyLongOperatorName}\left(x_1,x_2,x_3\right)=0$$

pathological.tex eq.2 (continuation, 3 lines)

A: original

$$S = a_1x_1+a_2x_2+a_3x_3+a_4x_4+a_5x_5+a_6x_6+a_7x_7+a_8x_8+a_9x_9+a_{10}x_{10}+a_{11}x_{11}+a_{12}x_{12}+a_{13}x_{13}+a_{14}x_{14}+a_{15}x_{15}+a_{16}x_{16}+a_{17}x_{17}+a_{18}x_{18}+a_{19}x_{19}+a_{20}x_{20}+a_{21}x_{21}+a_{22}x_{22}+a_{23}x_{23}+a_{24}x_{24}$$

D: math-layout

$$\begin{aligned} S = & a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + a_7x_7 + a_8x_8 + a_9x_9 + a_{10}x_{10} \\ & + a_{11}x_{11} + a_{12}x_{12} + a_{13}x_{13} + a_{14}x_{14} + a_{15}x_{15} + a_{16}x_{16} + a_{17}x_{17} \\ & + a_{18}x_{18} + a_{19}x_{19} + a_{20}x_{20} + a_{21}x_{21} + a_{22}x_{22} + a_{23}x_{23} + a_{24}x_{24} \end{aligned}$$

pathological.tex eq.4 (continuation-unanchored, 2 lines)

A: original

$$\text{MyLongOperatorName}\left(\text{MyLongOperatorName}\left(a_1+a_2+a_3+a_4\right)+\left(b_1+b_2+b_3+b_4\right)\right)=\left[c_1+c_2+c_3+c_4+c_5\right]$$

D: math-layout

$$\begin{aligned} & \text{MyLongOperatorName}\left(\text{MyLongOperatorName}\left(a_1+a_2+a_3+a_4\right)+\left(b_1+b_2+b_3+b_4\right)\right) \\ & = \left[c_1+c_2+c_3+c_4+c_5\right] + \{d_1+d_2+d_3\} \end{aligned}$$

pathological.tex eq.8 (continuation-unanchored, 2 lines)

A: original

$$\left(\alpha_1x_1+\alpha_2x_2+\alpha_3x_3+\alpha_4x_4+\alpha_5x_5+\alpha_6x_6+\alpha_7x_7+\alpha_8x_8+\alpha_9x_9+\alpha_{10}x_{10}+\alpha_{11}x_{11}+\alpha_{12}x_{12}+\alpha_{13}x_{13}+\alpha_{14}x_{14}+\alpha_{15}x_{15}+\alpha_{16}x_{16}+\alpha_{17}x_{17}+\alpha_{18}x_{18}\right)=1$$

D: math-layout

$$\begin{aligned} & \left(\alpha_1x_1 + \alpha_2x_2 + \alpha_3x_3 + \alpha_4x_4 + \alpha_5x_5 + \alpha_6x_6 + \alpha_7x_7 + \alpha_8x_8 + \alpha_9x_9 + \alpha_{10}x_{10} \right. \\ & \left. + \alpha_{11}x_{11} + \alpha_{12}x_{12} + \alpha_{13}x_{13} + \alpha_{14}x_{14} + \alpha_{15}x_{15} + \alpha_{16}x_{16} + \alpha_{17}x_{17} + \alpha_{18}x_{18}\right) = 1 \end{aligned}$$

residual-xgb.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$G(r) = L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{\text{XGB}}(x_i))^2 \geq \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{\text{lin}}(x_i))^2 + \frac{1}{n} \sum_{i=1}^n (\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i))^2$$

D: math-layout

$$\begin{aligned} G(r) &= L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{\text{XGB}}(x_i))^2 \\ &\geq \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{\text{lin}}(x_i))^2 + \frac{1}{n} \sum_{i=1}^n (\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i))^2 \end{aligned}$$

Baselines: autobreak (B) and breqn (C) vs math-layout (D)

bayes.tex eq.0

A: original

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D} \mid \vartheta) p(\vartheta) d\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta) p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)} [\prod_{i=1}^n p(y_i \mid x_i, \vartheta)]} + \frac{1}{2} \log \left(\frac{|\Sigma_0|}{|\Sigma_{\theta|\mathcal{D}}|} \right) + \text{KL}(q(\theta) \parallel p(\theta)) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta)$$

B: autobreak

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D} \mid \vartheta) p(\vartheta) d\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta) p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)} [\prod_{i=1}^n p(y_i \mid x_i, \vartheta)]} + \frac{1}{2} \log \left(\frac{|\Sigma_0|}{|\Sigma_{\theta|\mathcal{D}}|} \right) + \text{KL}(q(\theta) \parallel p(\theta)) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta)$$

C: breqn dmath

$$\begin{aligned} p(\theta \mid \mathcal{D}) &= \frac{p(\mathcal{D} \mid \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D} \mid \vartheta) p(\vartheta) d\vartheta} \\ &= \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta) p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)} [\prod_{i=1}^n p(y_i \mid x_i, \vartheta)]} + \frac{1}{2} \log \left(\frac{|\Sigma_0|}{|\Sigma_{\theta|\mathcal{D}}|} \right) + \text{KL}(q(\theta) \parallel p(\theta)) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta) - \frac{1}{2} \log \det \Sigma_{\theta|\mathcal{D}} \end{aligned}$$

D: math-layout

$$\begin{aligned} p(\theta \mid \mathcal{D}) &= \frac{p(\mathcal{D} \mid \theta) p(\theta)}{\int_{\Theta} p(\mathcal{D} \mid \vartheta) p(\vartheta) d\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta) p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)} [\prod_{i=1}^n p(y_i \mid x_i, \vartheta)]} \\ &\quad + \frac{1}{2} \log \left(\frac{|\Sigma_0|}{|\Sigma_{\theta|\mathcal{D}}|} \right) + \text{KL}(q(\theta) \parallel p(\theta)) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta) - \frac{1}{2} \log \det \Sigma_{\theta|\mathcal{D}} \end{aligned}$$

diffusion.tex eq.0

A: original

$$\mathbb{E} [\|x_{t+1} - x_t\|^2] = \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E} [\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2] + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2} \|\nabla_x \log p_t(x_t) - s_{\theta}(x_t, t)\|^2$$

B: autobreak

$$\mathbb{E} [\|x_{t+1} - x_t\|^2] = \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E} [\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2] + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2} \|\nabla_x \log p_t(x_t) - s_{\theta}(x_t, t)\|^2$$

C: breqn dmath

$$\mathbb{E} [\|x_{t+1} - x_t\|^2] = \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E} [\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2] + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2} \|\nabla_x \log p_t(x_t) - s_{\theta}(x_t, t)\|^2$$

D: math-layout

$$\begin{aligned} \mathbb{E} [\|x_{t+1} - x_t\|^2] &= \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E} [\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2] \\ &\quad + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2} \|\nabla_x \log p_t(x_t) - s_{\theta}(x_t, t)\|^2 \end{aligned}$$

esr.tex eq.0

A: original

$$\mathcal{L}(\theta) = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_\theta(x_i))^2 + \frac{1}{2} \log|\Sigma_\theta| + \frac{d}{2} \log(2\pi) + \lambda \sum_{j=1}^p |\theta_j| + \frac{n}{2} \log\left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right)$$

B: autobreak

$$\mathcal{L}(\theta) = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_\theta(x_i))^2 + \frac{1}{2} \log|\Sigma_\theta| + \frac{d}{2} \log(2\pi) + \lambda \sum_{j=1}^p |\theta_j| + \frac{n}{2} \log\left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right)$$

C: breqn dmath

$$\mathcal{L}(\theta) = \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_\theta(x_i))^2 + \frac{1}{2} \log|\Sigma_\theta| + \frac{d}{2} \log(2\pi) + \lambda \sum_{j=1}^p |\theta_j| + \frac{n}{2} \log\left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right)$$

D: math-layout

$$\begin{aligned} \mathcal{L}(\theta) = & \frac{1}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - f_\theta(x_i))^2 + \frac{1}{2} \log|\Sigma_\theta| \\ & + \frac{d}{2} \log(2\pi) + \lambda \sum_{j=1}^p |\theta_j| + \frac{n}{2} \log\left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2\right) \end{aligned}$$

kalman.tex eq.0

A: original

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} H_t^\top (H_t P_{t|t-1} H_t^\top + R_t)^{-1} H_t P_{t|t-1} + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t$$

B: autobreak

$$P_{t|t} = P_{t|t-1} - P_{t|t-1} H_t^\top (H_t P_{t|t-1} H_t^\top + R_t)^{-1} H_t P_{t|t-1} + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t$$

C: breqn dmath

$$\begin{aligned} P_{t|t} = & P_{t|t-1} \\ & - P_{t|t-1} H_t^\top (H_t P_{t|t-1} H_t^\top + R_t)^{-1} H_t P_{t|t-1} + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t \end{aligned}$$

D: math-layout

$$\begin{aligned} P_{t|t} = & P_{t|t-1} - P_{t|t-1} H_t^\top (H_t P_{t|t-1} H_t^\top + R_t)^{-1} H_t P_{t|t-1} \\ & + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t \end{aligned}$$

kkt.tex eq.0

A: original

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m (\max\{0, g_i(x)\})^2 + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^*\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho}$$

B: autobreak *not applicable: autobreak cannot parse this body (nested environment or \{\wedge\} delimiters).*

C: breqn dmath

$$\begin{aligned}\mathcal{L}(x, \lambda, \nu) = & f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m (\max\{0, g_i(x)\})^2 \\ & + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^\star\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho}\end{aligned}$$

D: math-layout

$$\begin{aligned}\mathcal{L}(x, \lambda, \nu) = & f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m (\max\{0, g_i(x)\})^2 \\ & + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^\star\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho}\end{aligned}$$

koopman.tex eq.0

A: original

$$\tilde{A} = \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y = (\Psi_X^\top \Psi_X)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon (\Psi_X^\top \Psi_X + \lambda I)^{-1} \Psi_X^\top \Psi_Y + \|\Psi_Y - \Psi_X \tilde{A}\|_F^2 + \lambda \|\tilde{A}\|_* + \|\Psi_X^\top \Psi_X - I\|_F$$

B: autobreak

$$\tilde{A} = \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y = (\Psi_X^\top \Psi_X)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon (\Psi_X^\top \Psi_X + \lambda I)^{-1} \Psi_X^\top \Psi_Y + \|\Psi_Y - \Psi_X \tilde{A}\|_F^2 + \lambda \|\tilde{A}\|_* + \|\Psi_X^\top \Psi_X - I\|_F$$

C: breqn dmath

$$\begin{aligned}\tilde{A} &= \tilde{U} \tilde{\Sigma} \tilde{V}^\top \\ &= \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top \\ &= \Psi_X^\dagger \Psi_Y \\ &= (\Psi_X^\top \Psi_X)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon (\Psi_X^\top \Psi_X + \lambda I)^{-1} \Psi_X^\top \Psi_Y + \|\Psi_Y - \Psi_X \tilde{A}\|_F^2 + \lambda \|\tilde{A}\|_* + \|\Psi_X^\top \Psi_X - I\|_F\end{aligned}$$

D: math-layout

$$\begin{aligned}\tilde{A} &= \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y \\ &= (\Psi_X^\top \Psi_X)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon (\Psi_X^\top \Psi_X + \lambda I)^{-1} \Psi_X^\top \Psi_Y + \|\Psi_Y - \Psi_X \tilde{A}\|_F^2 + \lambda \|\tilde{A}\|_* + \|\Psi_X^\top \Psi_X - I\|_F\end{aligned}$$

lyapunov.tex eq.0

A: original

$$\lambda_{\max} \left(A^\top P + PA + Q \right) \leq -\gamma \|x(t)\|^2 + \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, dt - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, ds - \delta \|x\| \|\dot{x}\|$$

B: autobreak

$$\lambda_{\max} \left(A^\top P + PA + Q \right) \leq -\gamma \|x(t)\|^2 + \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, dt - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, ds - \delta \|x\| \|\dot{x}\|$$

C: breqn dmath

$$\begin{aligned} \lambda_{\max} (A^\top P + PA + Q) &\leq -\gamma \|x(t)\|^2 + \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, dt \\ &\quad - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, ds - \delta \|x\| \|\dot{x}\| \end{aligned}$$

D: math-layout

$$\begin{aligned} \lambda_{\max} (A^\top P + PA + Q) &\leq -\gamma \|x(t)\|^2 + \limsup_{T \rightarrow \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, dt \\ &\quad - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, ds - \delta \|x\| \|\dot{x}\| \end{aligned}$$

matrix.tex eq.0

A: original

$$P = A (A^\top A + \lambda I)^{-1} A^\top = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} = U_r U_r^\top - (I - U_r U_r^\top) + 2U_r U_r^\top - (I - U_r U_r^\top) + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1}$$

B: autobreak *not applicable: autobreak cannot parse this body (nested environment or \{\wedge\} delimiters).*

C: breqn dmath

$$\begin{aligned} P &= A (A^\top A + \lambda I)^{-1} A^\top \\ &= \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \\ &= U_r U_r^\top - (I - U_r U_r^\top) + 2U_r U_r^\top - (I - U_r U_r^\top) + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} \end{aligned}$$

D: math-layout

$$\begin{aligned} P &= A (A^\top A + \lambda I)^{-1} A^\top = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \\ &= U_r U_r^\top - (I - U_r U_r^\top) + 2U_r U_r^\top - (I - U_r U_r^\top) + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} \end{aligned}$$

matrix.tex eq.1

A: original

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \quad A^\dagger = V_r \Sigma_r^{-1} U_r^\top = \sum_{k=1}^r \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} + \lambda \mathcal{N}(A)^\perp$$

B: autobreak *not applicable: autobreak cannot parse this body (nested environment or \{\wedge\} delimiters).*

C: breqn dmath

$$\begin{aligned} \mathcal{N}(A) &= \{x \\ &\in \mathbb{R}^n \\ &: Ax \\ &= 0\}, \quad A^\dagger \\ &= V_r \Sigma_r^{-1} U_r^\top \\ &= \sum_{k=1}^r \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} + \lambda \mathcal{N}(A)^\perp \end{aligned}$$

D: math-layout

$$\begin{aligned} \mathcal{N}(A) &= \{x \in \mathbb{R}^n : Ax = 0\}, \quad A^\dagger = V_r \Sigma_r^{-1} U_r^\top \\ &= \sum_{k=1}^r \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^r \sigma_k u_k^\top v_k - (I - U_r U_r^\top) A (A^\top A + \lambda I)^{-1} + \lambda \mathcal{N}(A)^\perp \end{aligned}$$

nash.tex eq.0

A: original

$$\sigma^\star \in \arg \max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j + \eta \sum_{i \in S} \sum_{j \in T} (\sigma_i \tau_j)^2 - \kappa \|\sigma - \sigma^\star\|_1 + \gamma \|\nabla_{\sigma} \mathbb{E}[u]\|^2 - \beta \sum_{i \in S}$$

B: autobreak

$$\sigma^\star \in \arg \max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j + \eta \sum_{i \in S} \sum_{j \in T} (\sigma_i \tau_j)^2 - \kappa \|\sigma - \sigma^\star\|_1 + \gamma \|\nabla_{\sigma} \mathbb{E}[u]\|^2 - \beta \sum_{i \in S}$$

C: breqn dmath

$$\begin{aligned} \sigma^\star \in \arg \max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j \\ + \eta \sum_{i \in S} \sum_{j \in T} (\sigma_i \tau_j)^2 - \kappa \|\sigma - \sigma^\star\|_1 + \gamma \|\nabla_{\sigma} \mathbb{E}[u]\|^2 - \beta \sum_{i \in S} \sigma_i^2 \end{aligned}$$

D: math-layout

$$\begin{aligned} \sigma^\star \in \arg \max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j \\ + \eta \sum_{i \in S} \sum_{j \in T} (\sigma_i \tau_j)^2 - \kappa \|\sigma - \sigma^\star\|_1 + \gamma \|\nabla_{\sigma} \mathbb{E}[u]\|^2 - \beta \sum_{i \in S} \sigma_i^2 \end{aligned}$$

network-flow.tex eq.0

A: original

$$\sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \quad \forall v \in V, \quad 0 \leq x_a \leq u_a, \quad \forall a \in A, \quad \sum_{v \in V} b_v = 0, \quad \sum_{a \in A} c_a x_a \leq C_{\max}, \quad x_a$$

B: autobreak

$$\sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \quad \forall v \in V, \quad 0 \leq x_a \leq u_a, \quad \forall a \in A, \quad \sum_{v \in V} b_v = 0, \quad \sum_{a \in A} c_a x_a \leq C_{\max}, \quad x_a$$

C: breqn dmath

$$\begin{aligned}
& \sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a \\
& = b_v, \quad \forall v \\
& \in V, \quad 0 \\
& \leq x_a \\
& \leq u_a, \quad \forall a \\
& \in A, \quad \sum_{v \in V} b_v \\
& = 0, \quad \sum_{a \in A} c_a x_a \\
& \leq C_{\max}, \quad x_a \\
& \geq 0
\end{aligned}$$

D: math-layout

$$\begin{aligned}
& \sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \quad \forall v \in V, \quad 0 \leq x_a \leq u_a, \quad \forall a \\
& \in A, \quad \sum_{v \in V} b_v = 0, \quad \sum_{a \in A} c_a x_a \leq C_{\max}, \quad x_a \geq 0
\end{aligned}$$

pathological.tex eq.1

A: original

$$f\left(g\left(h\left(k\left(x_1+x_2+x_3+x_4+x_5+x_6+x_7\right)\right)\right)\right)+\text{MyLongOperatorName}\left(x_1,x_2,x_3\right)=0$$

B: autobreak

$$f\left(g\left(h\left(k\left(x_1+x_2+x_3+x_4+x_5+x_6+x_7\right)\right)\right)\right)+\text{MyLongOperatorName}\left(x_1,x_2,x_3\right)=0$$

C: breqn dmath

$$f\left(g\left(h\left(k\left(x_1+x_2+x_3+x_4+x_5+x_6+x_7\right)\right)\right)\right)+\text{MyLongOperatorName}\left(x_1,x_2,x_3\right)=0$$

pathological.tex eq.2

A: original

$$S = a_1x_1+a_2x_2+a_3x_3+a_4x_4+a_5x_5+a_6x_6+a_7x_7+a_8x_8+a_9x_9+a_{10}x_{10}+a_{11}x_{11}+a_{12}x_{12}+a_{13}x_{13}+a_{14}x_{14}+a_{15}x_{15}+a_{16}x_{16}+a_{17}x_{17}$$

B: autobreak

$$S = a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + a_7x_7 + a_8x_8 + a_9x_9 + a_{10}x_{10} + a_{11}x_{11} + a_{12}x_{12} + a_{13}x_{13} + a_{14}x_{14} + a_{15}x_{15}$$

C: breqn dmath

$$\begin{aligned}
S = & a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + a_7x_7 + a_8x_8 + a_9x_9 + a_{10}x_{10} + a_{11}x_{11} + a_{12}x_{12} + a_{13}x_{13} \\
& + a_{14}x_{14} + a_{15}x_{15} + a_{16}x_{16} + a_{17}x_{17} + a_{18}x_{18} + a_{19}x_{19} + a_{20}x_{20} + a_{21}x_{21} + a_{22}x_{22} + a_{23}x_{23} + a_{24}x_{24}
\end{aligned}$$

D: math-layout

$$\begin{aligned}
S = & a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + a_7x_7 + a_8x_8 + a_9x_9 + a_{10}x_{10} \\
& + a_{11}x_{11} + a_{12}x_{12} + a_{13}x_{13} + a_{14}x_{14} + a_{15}x_{15} + a_{16}x_{16} + a_{17}x_{17} \\
& + a_{18}x_{18} + a_{19}x_{19} + a_{20}x_{20} + a_{21}x_{21} + a_{22}x_{22} + a_{23}x_{23} + a_{24}x_{24}
\end{aligned}$$

pathological.tex eq.4

A: original

MyLongOperatorName (MyLongOperatorName ($a_1 + a_2 + a_3 + a_4$) + ($b_1 + b_2 + b_3 + b_4$)) = [$c_1 + c_2 + c_3 + c_4 + c_5$]+{

B: autobreak *not applicable: autobreak cannot parse this body (nested environment or \{\} delimiters).*

C: breqn dmath

$$\text{MyLongOperatorName} \left(\text{MyLongOperatorName} (a_1 + a_2 + a_3 + a_4) + (b_1 + b_2 + b_3 + b_4) \right) \\ = [c_1 + c_2 + c_3 + c_4 + c_5] + \{d_1 + d_2 + d_3\}$$

D: math-layout

$$\text{MyLongOperatorName} \left(\text{MyLongOperatorName} (a_1 + a_2 + a_3 + a_4) + (b_1 + b_2 + b_3 + b_4) \right) \\ = [c_1 + c_2 + c_3 + c_4 + c_5] + \{d_1 + d_2 + d_3\}$$

pathological.tex eq.8

A: original

($\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 + \alpha_5 x_5 + \alpha_6 x_6 + \alpha_7 x_7 + \alpha_8 x_8 + \alpha_9 x_9 + \alpha_{10} x_{10} + \alpha_{11} x_{11} + \alpha_{12} x_{12} + \alpha_{13} x_{13} + \alpha_{14} x_{14} + \alpha_{15} x_{15} + \alpha_{16} x_{16} + \alpha_{17} x_{17} +$

B: autobreak

($\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 + \alpha_5 x_5 + \alpha_6 x_6 + \alpha_7 x_7 + \alpha_8 x_8 + \alpha_9 x_9 + \alpha_{10} x_{10} + \alpha_{11} x_{11} + \alpha_{12} x_{12} + \alpha_{13} x_{13} + \alpha_{14} x_{14} + \alpha_{15} x_{15} +$

C: breqn dmath

$$(\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 + \alpha_5 x_5 + \alpha_6 x_6 + \alpha_7 x_7 + \alpha_8 x_8 + \alpha_9 x_9 + \alpha_{10} x_{10} \\ + \alpha_{11} x_{11} + \alpha_{12} x_{12} + \alpha_{13} x_{13} + \alpha_{14} x_{14} + \alpha_{15} x_{15} + \alpha_{16} x_{16} + \alpha_{17} x_{17} + \alpha_{18} x_{18}) = 1$$

D: math-layout

$$(\alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \alpha_4 x_4 + \alpha_5 x_5 + \alpha_6 x_6 + \alpha_7 x_7 + \alpha_8 x_8 + \alpha_9 x_9 + \alpha_{10} x_{10} \\ + \alpha_{11} x_{11} + \alpha_{12} x_{12} + \alpha_{13} x_{13} + \alpha_{14} x_{14} + \alpha_{15} x_{15} + \alpha_{16} x_{16} + \alpha_{17} x_{17} + \alpha_{18} x_{18}) = 1$$

residual-xgb.tex eq.0

A: original

$$G(r) = L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{XGB}}(x_i) \right)^2 \geq \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{lin}}(x_i) \right)^2 + \frac{1}{n} \sum_{i=1}^n \left(\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i) \right)^2$$

B: autobreak

$$G(r) = L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{XGB}}(x_i) \right)^2 \geq \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{lin}}(x_i) \right)^2 + \frac{1}{n} \sum_{i=1}^n \left(\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i) \right)^2$$

C: breqn dmath

$$G(r) = L_{\text{null}} - L_{\text{XGB}} \\ = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{XGB}}(x_i) \right)^2 \\ \geq \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{lin}}(x_i) \right)^2 + \frac{1}{n} \sum_{i=1}^n \left(\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i) \right)^2$$

D: math-layout

$$\begin{aligned} G(r) &= L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{XGB}}(x_i) \right)^2 \\ &\geq \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{f}_{\text{lin}}(x_i) \right)^2 + \frac{1}{n} \sum_{i=1}^n \left(\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i) \right)^2 \end{aligned}$$